

Late Breaking Results: Modern Automatic PCB Placement with Complex Constraints

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ABSTRACT

Existing printed circuit board (PCB) placement often fails to address complex constraints (e.g., diverse wire widths and intricate spacing rules) arising from heterogeneous components in modern designs. Manual placement requires expertise and is time-consuming. Thus, automated PCB placement is desired for large-scale, complex designs considering irregular component shapes, clearance conditions, wire areas, power circuit flow, and routability. This paper proposes an efficient force-directed global placement followed by legalization to handle these constraints. We derive power circuit anchor points to guide global placement when considering the power circuit flow and present a simulated annealing-based pad alignment method to handle complex spacing constraints. After global placement, we perform window-based legalization to remove component overlaps considering various constraints. Experimental results show our placer's superior routability and efficiency with complex constraints.

1 INTRODUCTION

Existing automatic printed circuit board (PCB) placement often fails to address complex constraints (e.g., diverse wire widths and intricate spacing rules) arising from heterogeneous components in modern designs. Manual placement requires expertise and is time-consuming. Thus, automated PCB placement is desired for large-scale, complex designs considering irregular component shapes, clearance conditions, wire areas, power circuit flows, and routability. Figure 1 illustrates a PCB placement on a two-layer board with various component sizes considering the above complex constraints.

For automatic PCB placement, the work [1] presented a force-directed method and a computational procedure for solving the resulting system of equations. The work [2] described a rule-based technique based on bus structure classification. Cheng et al. [3] proposed a maximum-margin formulation for net separation.

Modern automatic PCB placement becomes challenging because various complex constraints must be considered. Advanced PCBs often contain special constraints, such as irregular component shapes, clearance conditions, and wire areas. The irregular component shape means that the contour of each component is not rectangular. Clearance conditions refer to the minimum distance between pads based on the clearance table given by the design rule. Each routing layer has its preferred wire width. As a result, we estimate the *wire area* calculated by $wirelength \times preferred\ wire\ width$, not only wirelength. Most previous works consider only wirelength [4–6], not wire areas.

The power circuit flow is a crucial measure in evaluating the placement quality, where components of a circuit need to follow the circuit flow for their placement to achieve high performance. In [7], circuit flow constraints were managed to streamline the datapath in mixed-size circuits with macros and standard cells. To enhance routability, researchers proposed approaches to minimize total wirelength [1, 2] and crossing nets [3].

To address the above challenges, we propose a novel force-directed placement algorithm for PCB designs with heterogeneous components. This algorithm includes four key techniques: (1) an efficient force-directed global placement followed by legalization to handle complex constraints, (2) a simulated annealing-based pad alignment method to address spacing constraints, and (3) utilization of anchor points from the power circuit flow to guide global placement, and (4) window-based legalization to remove component overlaps considering various constraints. Experimental results demonstrate our placer's superior performance in routability and runtime under diverse constraints.

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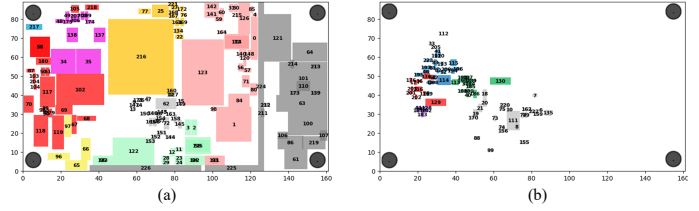


Figure 1: A double-sided PCB placement. Components with the same color belong to the same class. (a) Front-side placement. (b) Back-side placement.

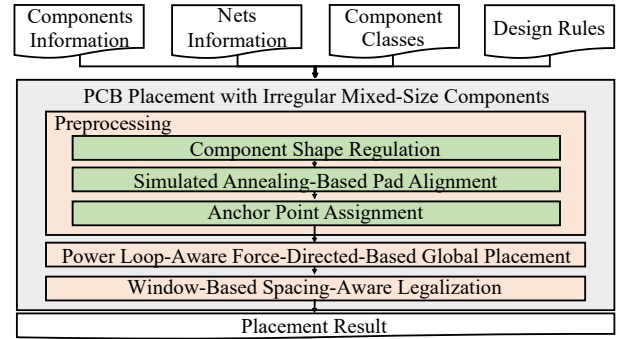


Figure 2: Our PCB placement flow.

2 PROBLEM FORMULATION

PROBLEM 1 (AUTOMATIC PCB PLACEMENT WITH COMPLEX CONSTRAINTS PROBLEM). Given a set of movable and fixed components, a netlist, and spacing (clearance) constraints, the objective is to place all the components on the board to maximize the routability and minimize the total wire area while no design rules are violated.

3 PLACEMENT FLOW

Figure 2 shows the overall flow of our proposed algorithm, where the three major stages are detailed in the following subsections.

3.1 Preprocessing

3.1.1 Component Shape Regulation. During preprocessing, we approximate each component with a minimum bounding box to reduce irregularity. To address complex spacing constraints, components of similar dimensions and the same functionality are grouped into clusters for the *simulated annealing-based pad alignment*. To enhance optimization for global placement, we position anchor points based on the power circuit flow on the board in the *anchor point assignment*.

3.1.2 Simulated Annealing-Based Pad Alignment. As surface mount device (SMD) components in the same component classes are grouped to form the same functionality, we propose a simulated annealing (SA)-based pad alignment method to minimize crossing and wire area within the components of the groups. For each group, we build an $n \times n$ grid graph, and the vertices of the grid graph represent the possible locations for the pad of SMD components, where n is the total number of pads of the components in this group. The grid graph ensures that the pads are aligned to each other. There are two operations for components in SA, Movement and Rotation. The cost function is to minimize the wire area and the number of crossings. After the SA process, we set the spacing for two adjacent rows (columns) to the maximum of the minimum pad spacings between any two

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Table 1: Experimental result.

Benchmark	# Components	# Nets	ePlace		Ours w/ only <i>SA-based Pad Alignment</i>		Ours w/ only <i>Anchor Point Assignment</i>		Ours with both techniques	
			Routability	Runtime (s)	Routability	Runtime (s)	Routability	Runtime (s)	Routability	Runtime (s)
PCB1	225	426	79.81%	1512.50	80.99%	1919.56	85.45%	2295.71	92.72%	1365.00
PCB2	219	416	80.77%	1538.83	80.77%	1926.65	85.10%	2372.19	92.30%	1328.33
PCB3	204	386	84.46%	1539.80	83.42%	1888.51	86.52%	2138.78	94.82%	1147.72
PCB4	211	402	82.34%	1660.39	83.33%	1941.20	85.82%	2222.30	94.02%	1168.00
PCB5	190	362	84.25%	1441.62	83.43%	1762.77	86.46%	2083.52	95.30%	1020.12
PCB6	205	392	82.91%	1565.58	83.42%	1861.03	85.46%	2266.34	93.36%	1081.67
PCB7	198	376	84.04%	1569.29	83.24%	1889.64	86.44%	2151.51	95.21%	1073.85
Average	-	-	82.65%	1546.86	82.66%	1884.20	85.89%	2218.62	93.96%	1,169.24
Comp.	-	-	1	1	1.0001	1.2181	1.0392	1.4343	1.1368	0.7559

pads of the adjacent rows (columns). Thus, we can obtain a regular layout with a minimized wire area and number of crossings.

3.1.3 Anchor Point Assignment. The power circuit flow is conducted in a line if there is no limited region on the PCB layout. However, the board's outline is fixed; hence, the flow could be bent. To fit the placement region, we establish several anchor points based on the power circuit flow to guide the overall layout.

First, we generate anchor points along the path for each component class based on their total area ratio. The path length is set to the sum of the maximum heights or widths of all components. Then, we curve the path into a wave to fit the board shape. Finally, we allocate the anchor points on the board and place components according to the corresponding component class. Figure 3 shows an example with five component classes, where the areas of the classes are 100, 150, 50, 75, and 25 mm^2 . As the I/O components are given and fixed inside the board, we naturally curve the path to the wave and align the path's endpoints with the locations of the I/O components.

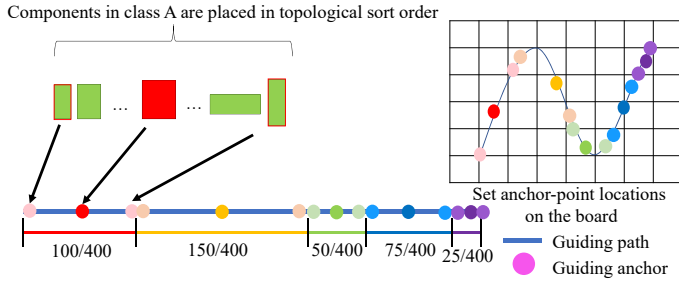


Figure 3: Anchor points from the power circuit flow, where 100 is the total area of the class-A components and 400 is the total component area.

3.2 Power Loop-Aware Force-Directed-Based Global Placement

We use two forces on the component to position the component and the relative position of the component.

3.2.1 Attractive force. To enhance the signal flow smoothness and maintain the functional group integrity, we position components belonging to the same net, component class, and the components of the same power circuit flow closer to each other. Therefore, we formulate attractive forces for component C_i on the x-axis as follows:

$$F_{x_i} = \sum w_{ij} \times \frac{d_x}{d}, \quad (1)$$

where x_i is the x-axis location of the component C_i , and w_{ij} is the weight between components C_i and C_j according to the net information, component classes, power circuit flow, and anchor point, $d_x = x_i - x'_i$, $d = \sqrt{d_x^2 + d_y^2}$, and $x'_i = \frac{\sum_j w_{ij} x_j}{\sum_j w_{ij}}$.

3.2.2 Repulsive force. To consider the overlap between components and the clearance of pads, we introduce repulsive force component i on the x-axis as follows:

$$F_{x_i} = \sum w_{ij} \times \frac{a_j}{a_i + a_j} \times l_{x_{ij}}, \quad (2)$$

where a_i is the area of component C_i , and $l_{x_{ij}}$ is the overlapping length along the x-axis between components C_i and C_j .

3.3 Window-Based Spacing-Aware Legalization

After global placement, we aim to eliminate overlaps between components in this stage. We perform window-based legalization to determine the location of each component. We begin by sorting the components, prioritizing those closest to the input on the board, and moving towards the output. To legalize the power components first, we legalize non-SMD components before the SMD ones. Our process establishes a search area for each component, with the width (height) of the search region being α times the component's width (height). In our experiment, α is set to 2.5. Starting from the center of the search area, we search outward in a spiral pattern to find a legal position. We choose the position closest to the center point with no spacing violation to place the component.

4 EXPERIMENTAL RESULTS

Our proposed algorithm was implemented in C++, and all experiments were conducted on an Intel Xeon Silver 4215R Linux machine with a 3.20 GHz CPU and 200 GB memory. Our benchmarks were from industrial cases. The routability was estimated from an industrial router, which is 11.5% better than the result of FreeRouting [8] on PCB1.

Table 1 gives the comparison of the routability and runtime (in seconds) between our proposed algorithm and ePlace [9]. The ePlace routability is only 87.96% of ours, and its runtime is 32.3% longer. Our placer achieves the best result for all cases. *Anchor Point Assignment* contributes a 12% improvement, while *SA-based Pad Alignment* contributes 8.6% to routability. The experimental results demonstrate the superiority of our algorithm.

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